Evaluation and Management of Tuberous Sclerosis Complex (TSC)

Initial Diagnostic Workup

The initial evaluation of a patient suspected to have TSC includes the following: 1. **Cranial computed tomography (CT) or cranial magnetic resonance imaging (MRI)** 2. **Renal ultrasonography** (or MRI or CT) 3. **Electrocardiography**

Additional tests are recommended based on clinical presentation and patient age: - **Electroencephalography (EEG)** is valuable if seizures are suspected and for ongoing seizure management. - In **children**, standardized **cognitive and behavioral assessments** should be performed at diagnosis, in response to behavioral changes or declining performance, and at regular, individualized intervals. - In **adult women**, **high-resolution chest CT** is recommended to screen for lymphangioleiomyomatosis (LAM). - **Echocardiography** is indicated if cardiac symptoms are present.

Periodic Surveillance After Diagnosis

After diagnosis, patients should undergo regular surveillance to monitor for new lesions or changes in existing ones: - **Cranial CT or MRI** every 1 to 3 years in children to detect subependymal giant cell astrocytomas (SEGAs) before they become symptomatic or locally invasive. - **Renal ultrasonography** every 1 to 3 years, depending on clinical suspicion and prior findings. If large or numerous renal tumors are present, **renal CT or MRI** is preferred for more detailed assessment. - **Echocardiography** and **chest CT** are performed if cardiac or pulmonary symptoms develop, respectively.

Molecular Diagnosis

Genetic testing can be useful in: - Confirming diagnosis in individuals who do not meet clinical criteria for definite TSC. - Providing information for **genetic counseling**. - Enabling **prenatal diagnosis**.

However, genetic testing has limitations: - Approximately **15% of TSC patients** have no identifiable mutation in *TSC1* or *TSC2* despite extensive analysis. - *TSC1* and *TSC2* are large genes with mutations potentially occurring anywhere in the sequence. - Standard sequencing may miss **large deletions**, requiring additional analytical methods. - **Somatic mosaicism** can complicate mutation detection, leading to false-negative or inconclusive results.

Differential Diagnosis

Box 141-1: Differential Diagnosis of Hypomelanotic Macules

- **Polygonal or ash-leaf-shaped** - Nevus depigmentosus - Piebaldism - Vitiligo - MEN1 hypomelanotic macules - Nevus anemicus
- **Confetti-like** - Idiopathic guttate hypomelanosis (adults) - MEN1 confetti lesions
- **Associated with poliosis** - Waardenburg syndrome
- Piebaldism
- Vitiligo - Vogt-Koyanagi-Harada syndrome
- MEN1 hypomelanotic macules
- Nevus anemicus - Segmental pigment mosaicism
  MEN1 = multiple endocrine neoplasia type 1

Box 141-2: Differential Diagnosis of Angiofibromas

- Fibrous papules
- Multiple endocrine neoplasia type 1 (MEN1) angiofibromas
- Birt-Hogg-Dubé syndrome angiofibromas
- Trichoepitheliomas

- Fibrofolliculomas/trichodiscomas
- Tricholemmomas
- Syringomas
- Dermal melanocytic nevi
- Acne vulgaris
- Rosacea

Box 141-3: Differential Diagnosis of Ungual Fibroma

Most Likely

Periungual: - Acral fibrokeratoma - Superficial acral fibromyxoma - Epidermoid cyst - Pseudomyxoid cyst - Wart - Pyogenic granuloma - Juvenile xanthogranuloma

Subungual: - Acral fibrokeratoma - Superficial acral fibromyxoma - Wart - Subungual exostosis - Subungual horn - Familial retinoblastoma - Pyogenic granuloma - Onychomycosis - Psoriasis

Always Rule Out (Subungual)

- Squamous cell carcinoma
- Bowen disease
- Amelanotic melanoma

Complications

None of the skin lesions in TSC are at risk for malignant transformation. However, they can cause significant **cosmetic concerns**, leading to **social isolation** and **impaired self-esteem** in affected individuals.